Desenvolvimento Mobile - Android

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Sumário

- Plataforma
- Componentes
- Serviços
- Ferramentas

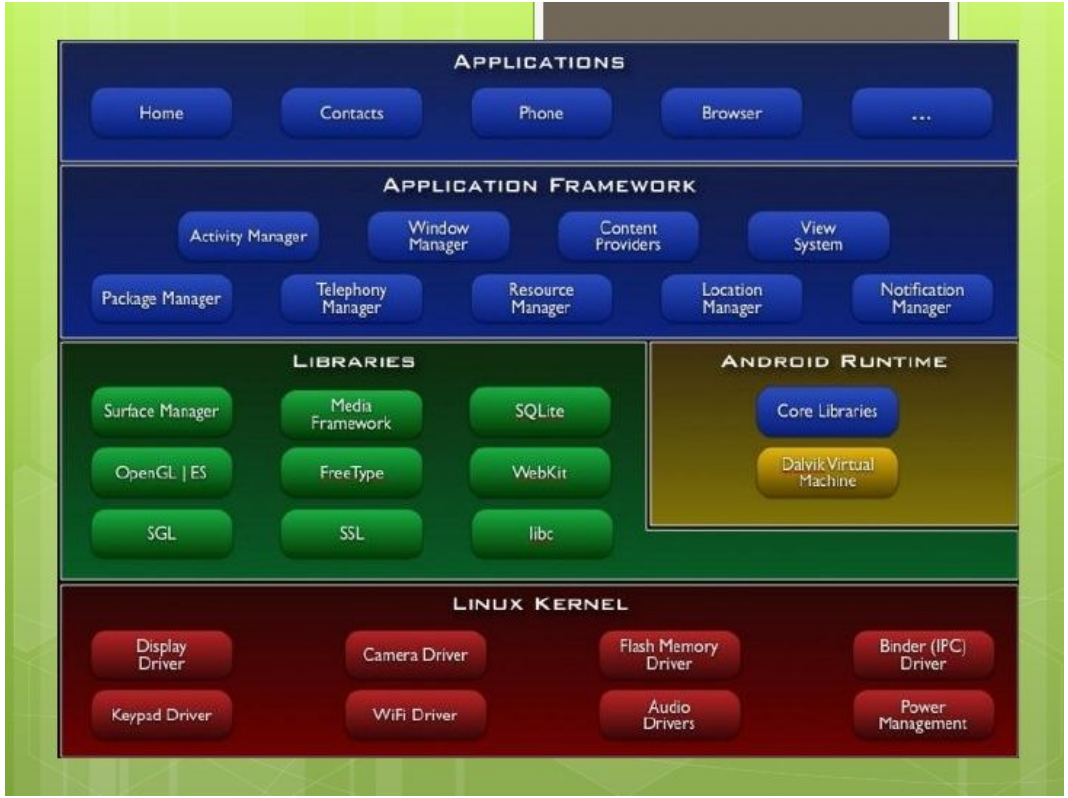
O que é?

- Sistema Operacional para dispositivos móveis (celulares, carros, smart-tvs, etc)
- Seu core é com base no kernel do Linux
- Comprada e desenvolvido pelo Google e a OHA (Open Handset Alliance)

O consórcio

中国移动通信
CHINA MOBILE
ASUS
Rock Solid · Heart Touching
SoftBank
T-Mobile
AKM
Google
SiRF
Synaptics
Designing The Future
KDDI
GARMIN
Sprint
TELECOM
esmertec
Audience
QUALCOMM
TELECA
NTT
DoCoMo
htc
smart mobility
MOTOROLA
TOSHIBA
Ascender Corporation
ebay
ARM
SkyPoP
HUAWEI
LG Electronics
SAMSUNG
Sony Ericsson
Telefonica
vodafone
NUANCE
pv
ERICSSON
TAKING YOU FORWARD
BROADCOM.
OMRON
ATHEROS
Living image
noser
WIND RIVER
TEXAS INSTRUMENTS
BORQS
Aplix
Corporation
SONiVOX
sound that rocks.
MARVELL
intel
NVIDIA.
tat

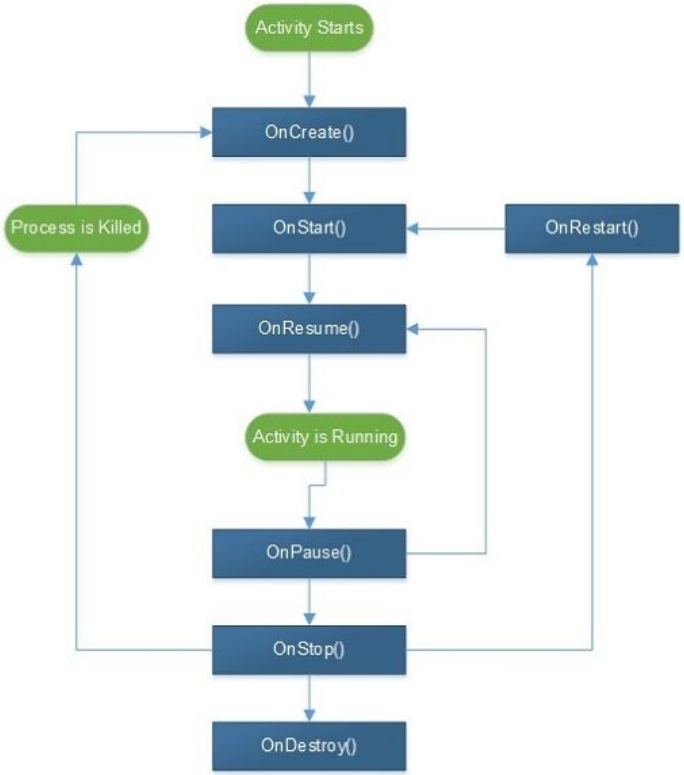
Plataforma


APPLICATIONS
Home
Contacts
Phone
Browser
...
APPLICATION FRAMEWORK
Activity Manager
Window Manager
Content Providers
View System
Package Manager
Telephony Manager
Resource Manager
Location Manager
Notification Manager
LIBRARIES
Surface Manager
Media Framework
SQLite
OpenGL | ES
FreeType
WebKit
SGL
SSL
libc
ANDROID RUNTIME
Core Libraries
Dalvik Virtual Machine
LINUX KERNEL
Display Driver
Camera Driver
Flash Memory Driver
Binder (IPC) Driver
Keypad Driver
WiFi Driver
Audio Drivers
Power Management

O que podemos criar ?

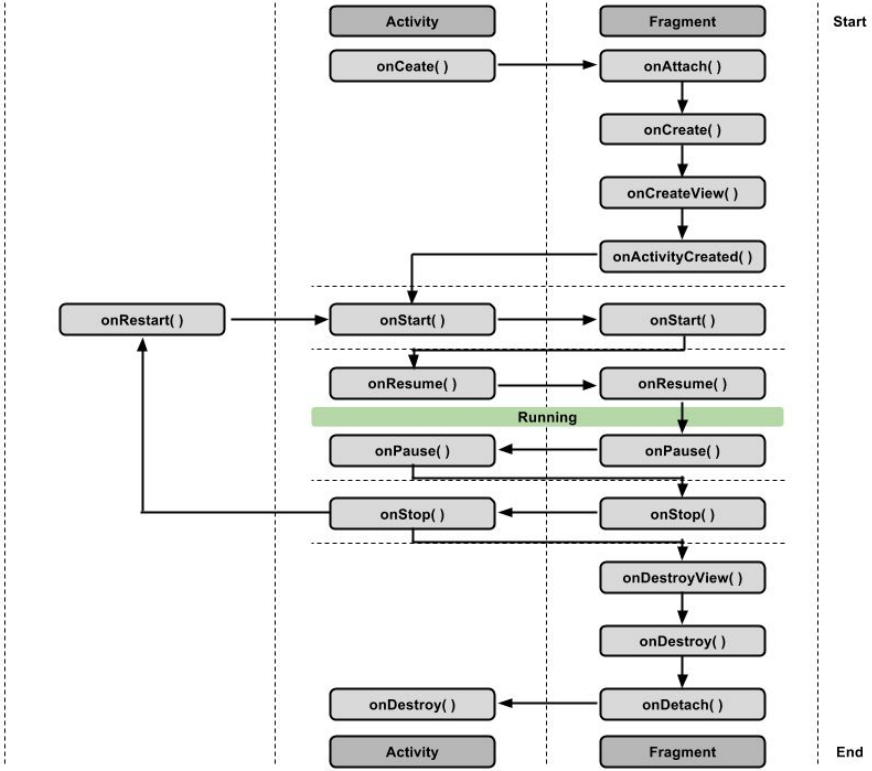
- Aplicativos
  - Activity
  - Fragments
- Services
- Libs (Ex.: AAR)

Activity


Activity Starts
OnCreate()
Process is Killed
OnStart()
OnRestart()
OnResume()
Activity is Running
OnPause()
OnStop()
OnDestroy()

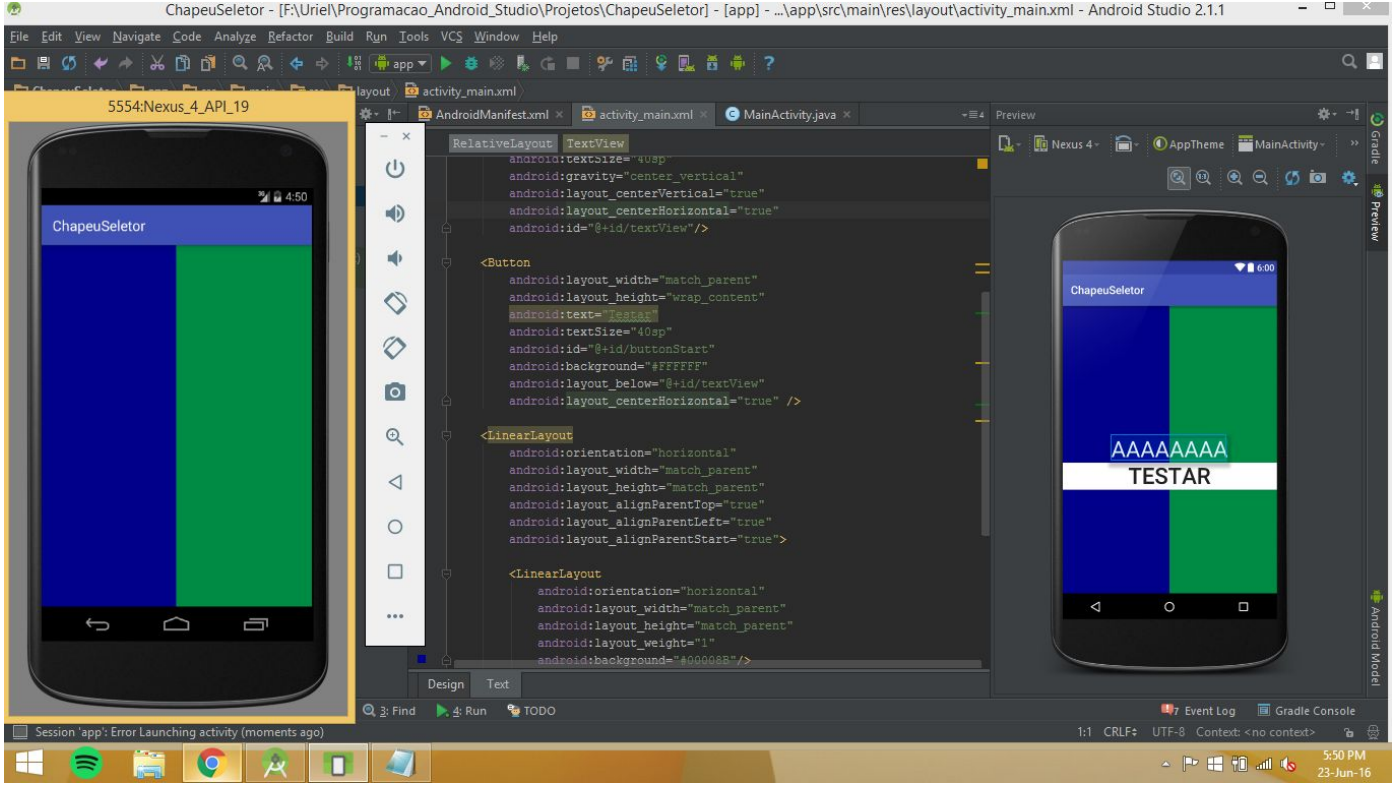
Activity

Fragments


Activity
Fragment
Start
onCeate( )
onAttach( )
onCreate( )
onCreateView( )
onActivityCreated( )
onRestart( )
onStart( )
onStart( )
onResume( )
onResume( )
Running
onPause( )
onPause( )
onStop( )
onStop( )
onDestroyView( )
onDestroy( )
onDestroy( )
onDetach( )
Activity
Fragment
End

22:50
DialogFragmentExample
Simple Dialog
Lorem ipsum dolor sit amet, eam at inermis eligendi, explicari referrentur nec et, no quo suavitate consequat. Pri hinc gloriatur in, graeci voluptatibus sea no..
Dismiss

Layout no Android



AndroidManifest.xml

Android

- app
  - manifests
    - AndroidManifest.xml
  - java
    - com.example.whatever.photoexamp
      - MainActivity
    - com.example.whatever.photoexamp
    - com.example.whatever.photoexamp
  - res
- Gradle Scripts

activity_main.xml × | MainActivity.java × | AndroidManifest.xml ×

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.whatever.photoexamplenew">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="PhotoExampleNew"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/AppTheme">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Gradle

app - Getting Started with CBM - [~/workspace/android/GettingStartedwithCBM]

GettingStartedwithCBM > app > build.gradle

Android

- app
  - manifests
  - java
    - com.couchbase.example.gettingstartedw
      - MainActivity
    - com.couchbase.example.gettingstartedw
    - com.couchbase.example.gettingstartedw
  - res
- Gradle Scripts
  - build.gradle (Project: GettingStartedwithCBM
  - build.gradle (Module: app)
  - gradle-wrapper.properties (Version)
  - proguard-rules.pro (ProGuard s for app)
  - gradle.properties (Project Properties
  - settings.gradle (Project Settings)
  - local.properties (SDK Location)

activity_main.xml | MainActivity.java | app

```gradle
apply plugin: 'com.android.application'

android {
    compileSdkVersion 23
    buildToolsVersion "23.0.3"

    defaultConfig {
        applicationId "com.couchbase.example.gettingstartedwithcbm"
        minSdkVersion 10
        targetSdkVersion 23
        versionCode 1
        versionName "1.0"
    }
    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
        }
    }
}

dependencies {
    compile fileTree(dir: 'libs', include: ['*.jar'])
    testCompile 'junit:junit:4.12'
    compile 'com.android.support:appcompat-v7:23.4.0'
}
```

versão 23 = Android 6.0

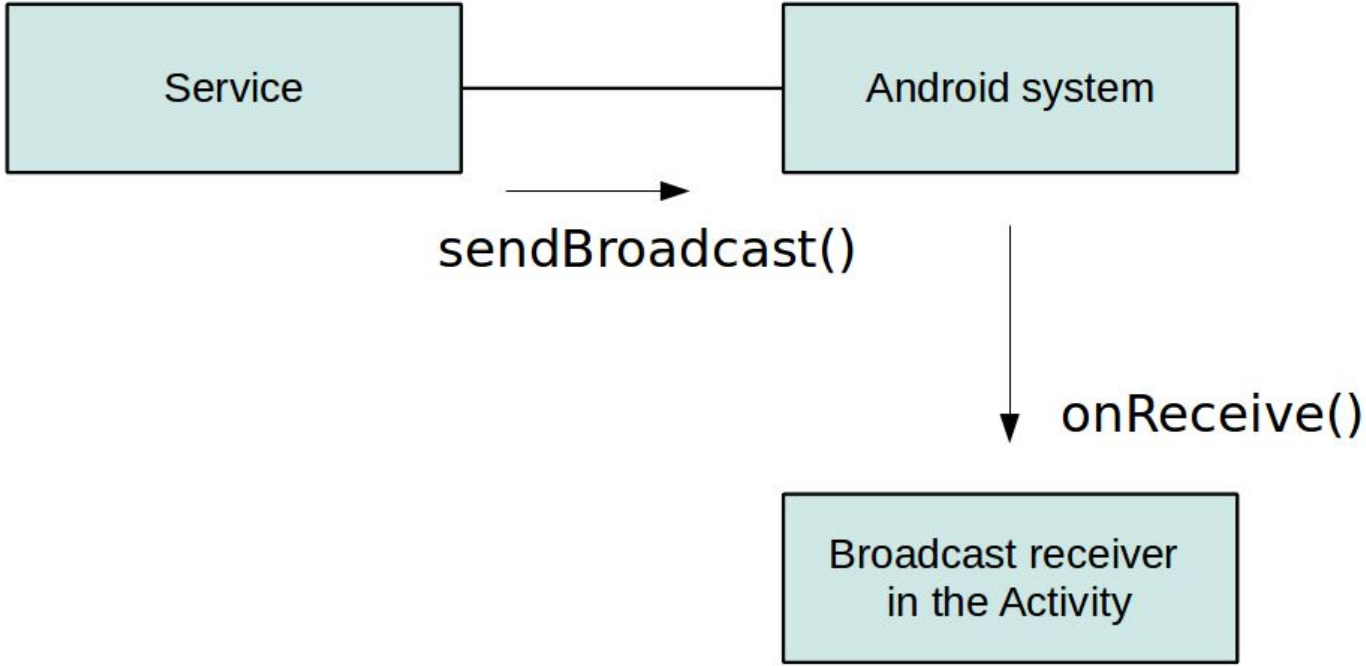
Versões

Code name	Version number	Initial release date	API level	Security patches[1]
(No codename)[2]	1.0	September 23, 2008	1	Unsupported
(Internally known as "Petit Four")[2]	1.1	February 9, 2009	2	Unsupported
Cupcake	1.5	April 27, 2009	3	Unsupported
Donut[3]	1.6	September 15, 2009	4	Unsupported
Eclair[4]	2.0 – 2.1	October 26, 2009	5 – 7	Unsupported
Froyo[5]	2.2 – 2.2.3	May 20, 2010	8	Unsupported
Gingerbread[6]	2.3 – 2.3.7	December 6, 2010	9 – 10	Unsupported
Honeycomb[7]	3.0 – 3.2.6	February 22, 2011	11 – 13	Unsupported
Ice Cream Sandwich[8]	4.0 – 4.0.4	October 18, 2011	14 – 15	Unsupported
Jelly Bean[9]	4.1 – 4.3.1	July 9, 2012	16 – 18	Unsupported
KitKat[10]	4.4 – 4.4.4	October 31, 2013	19 – 20	Supported;[11] See clarification
Lollipop[12]	5.0 – 5.1.1	November 12, 2014	21 – 22	Supported
Marshmallow[13]	6.0 – 6.0.1	October 5, 2015	23	Supported
Nougat[14]	7.0 – 7.1.2	August 22, 2016	24 – 25	Supported
Oreo	**8.0**	**August 21, 2017**	**26**	**Supported**

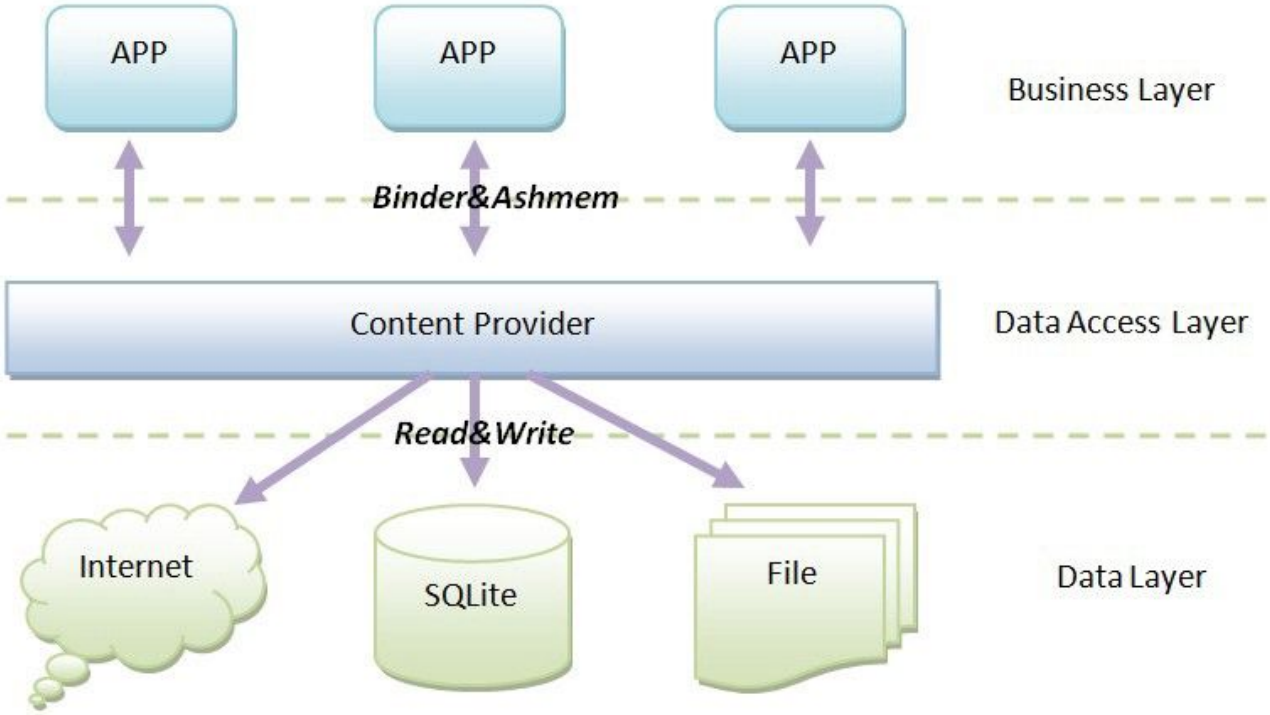
Componentes

- Broadcast Receiver
- Content Provider
- Notifications

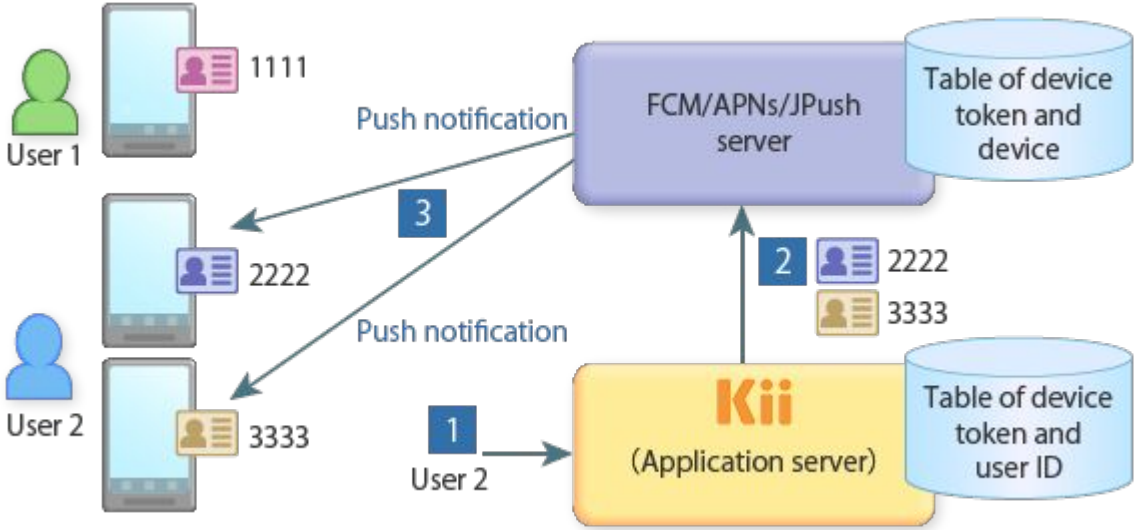
Broadcast Receiver



Content Provider



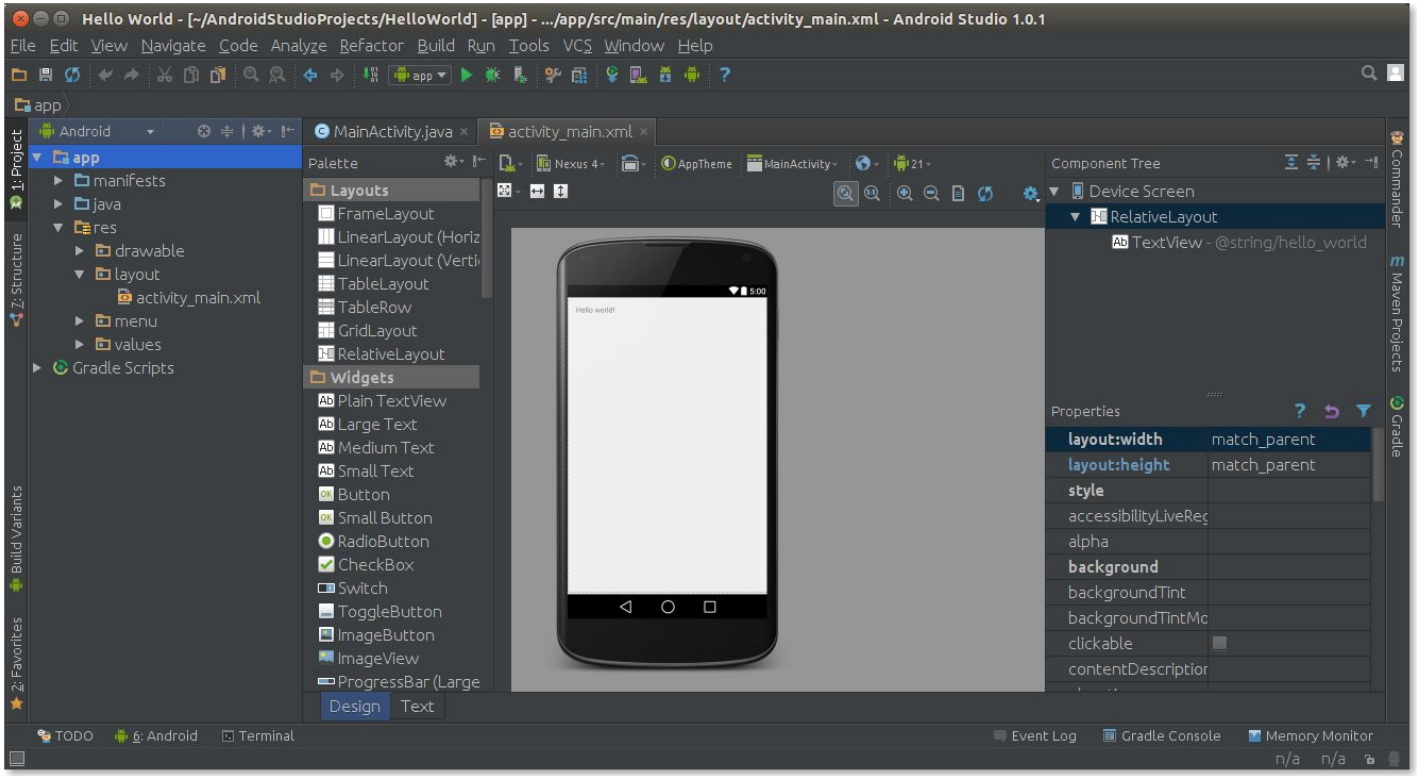
Notifications



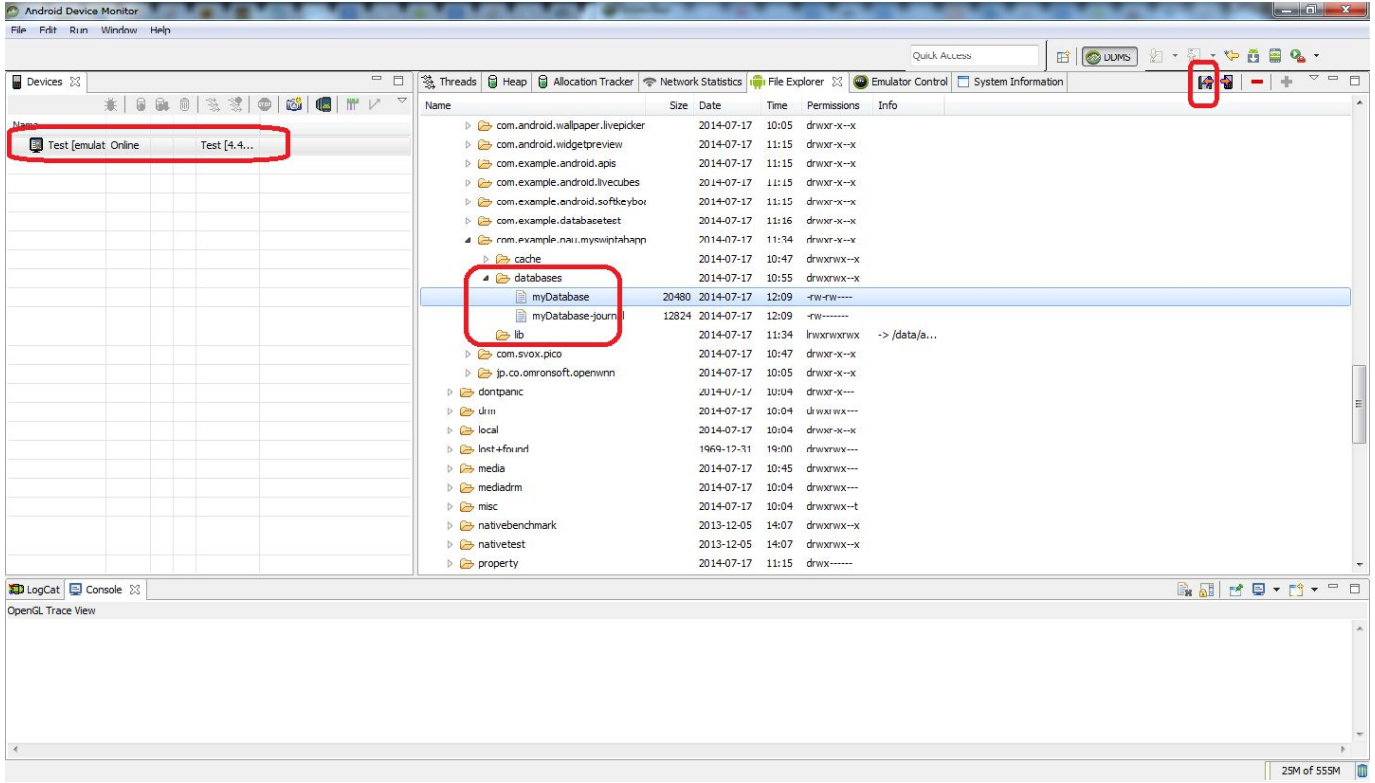
Notifications

Ferramentas

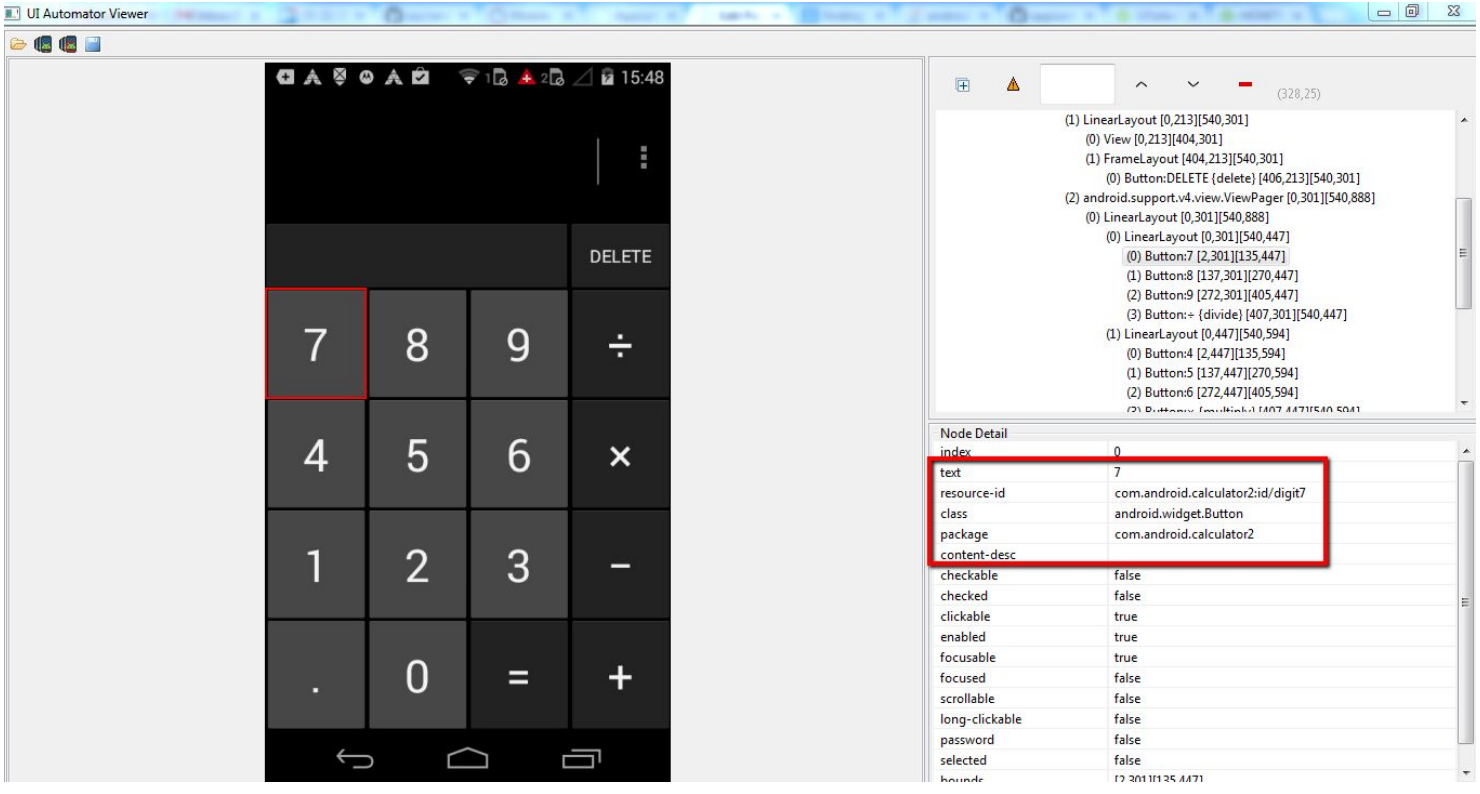
IDE


Hello World - [~/AndroidStudioProjects/HelloWorld] - [app] - .../app/src/main/res/layout/activity_main.xml - Android Studio 1.0.1
File Edit View Navigate Code Analyze Refactor Build Run Tools VCS Window Help
app
Android
app
manifests
java
res
drawable
layout
activity_main.xml
menu
values
Gradle Scripts
MainActivity.java
activity_main.xml
Palette
Layouts
FrameLayout
LinearLayout (Horiz
LinearLayout (Verti
TableLayout
TableRow
GridLayout
RelativeLayout
Widgets
Plain TextView
Large Text
Medium Text
Small Text
Button
Small Button
RadioButton
CheckBox
Switch
ToggleButton
ImageButton
ImageView
ProgressBar (Large
Nexus 4
AppTheme
MainActivity
Hello world!
Component Tree
Device Screen
RelativeLayout
TextView - @string/hello_world
Properties
layout:width
match_parent
layout:height
match_parent
style
accessibilityLiveReg
alpha
background
backgroundTint
backgroundTintMo
clickable
contentDescriptior
Design
Text
TODO
6: Android
Terminal
Event Log
Gradle Console
Memory Monitor
1: Project
7: Structure
Build Variants
2: Favorites
Commander
Maven Projects
Gradle

Ferramentas – Android Device Monitor



Vysor

You can now view the edit history of any user's Play Store reviews
ANDROID AUTHORITY

UiAutomation Viewer